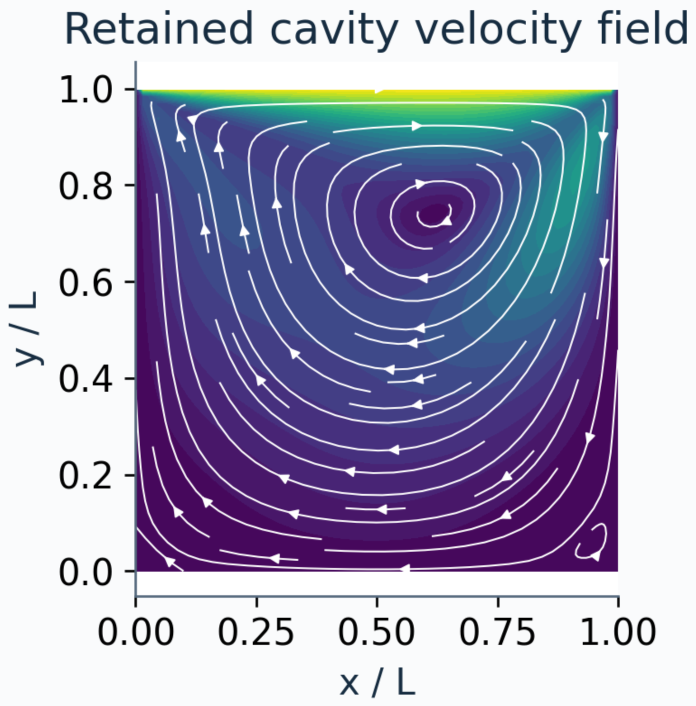
AI-assisted scientific software

From a proposed implementation to a defensible scientific result

Can you trust a plausible flow field?

A worked investigation of derivatives, conservation and reproducibility using the Re = 100 cavity.



Learning objective: write a contract, challenge the code, and defend a bounded claim.

Four questions before accepting a result

Each question requires a different kind of evidence.

Execution	**Does the program run?** Imports, finite outputs, expected shapes.
Verification	**Does it solve the equations correctly?** A known solution and a mesh-convergence study.
Validation	**Does the model represent the physical system?** Independent physical or experimental reference data.
Reproducibility	**Can the result be reconstructed?** Identified inputs, environment, commands and outputs.

This lab verifies a post-processor; the cavity check audits consistency with a retained field.

Make the scientific contract executable

Define semantics before asking a person or an agent to implement the diagnostic.

Contract item	Explicit choice	Why it matters
Coordinates	u[j, i] = u(y[j], x[i])	Axis swaps can preserve array shape.
Quantities	Divergence and out-of-plane vorticity	Signs determine physical interpretation.
Support	Exclude two layers for the interior audit	Boundary stencils have different support.
Precision	Float64; fixed tolerances	Small residuals depend on numerical precision.
Identity	Exact SHA-256 of the cavity archive	A changed input invalidates the record.
Decision	All seven gates must pass	One failure rejects the current claim.

Deliverable: SCIENTIFIC_SPEC.md plus a machine-readable acceptance record.

Axes and signs are part of the mathematics

Cartesian coordinates; u is the x-velocity and v is the y-velocity.

$$\nabla\cdot\mathbf{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}$$

$$\omega_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$$

Positive vorticity means counterclockwise rotation in the x-y plane.

axis=1 differentiates along x.
axis=0 differentiates along y.
Coordinates supply the spacing.

$$\left.\frac{\partial f}{\partial x}\right|_{j,i} = \frac{f_{j,i+1} - f_{j,i-1}}{2\Delta x} + \mathcal{O}(\Delta x^2)$$

The centered interior stencil has second-order truncation error on a uniform grid.

Construct a field with a known answer

The streamfunction makes incompressibility an analytic identity.

$$\psi = \sin^2(\pi x)\sin^2(\pi y), \qquad (x, y) \in [0, 1]^2$$

$$u = \psi_y, \qquad v = -\psi_x, \qquad \nabla \cdot \mathbf{u} = 0$$

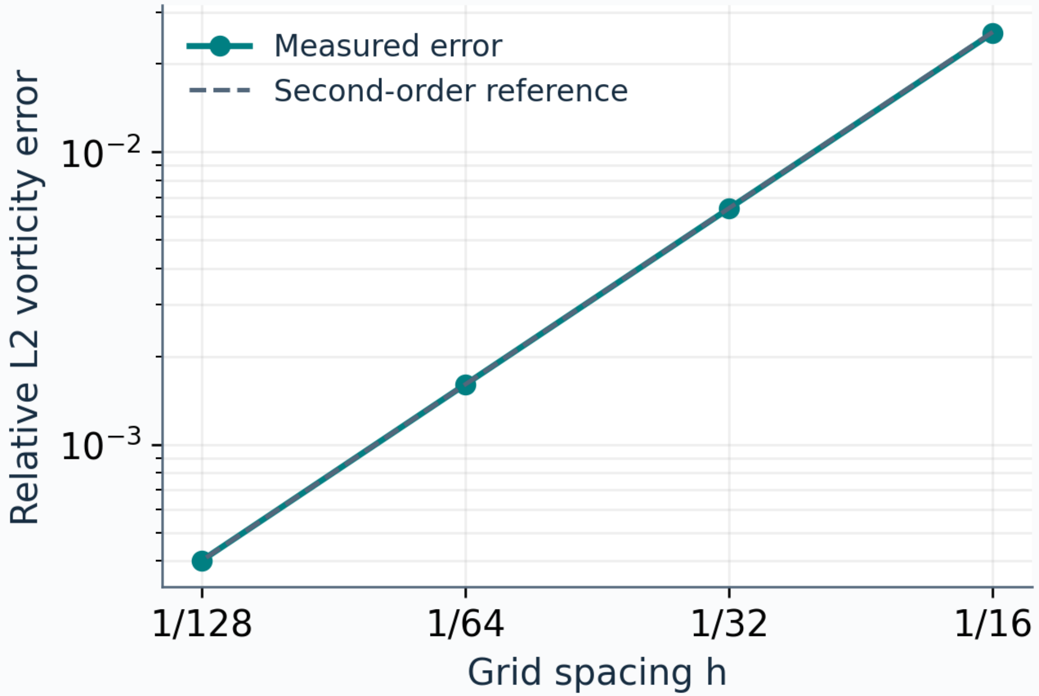
$$\omega_z = -\nabla^2 \psi = -2\pi^2 [\cos(2\pi x)\sin^2(\pi y) + \sin^2(\pi x)\cos(2\pi y)]$$

All wall velocities vanish. Sample the analytic expressions directly; compare numerical derivatives with the analytic vorticity.

A zero divergence residual alone cannot establish that vorticity is correct.

Demonstrate accuracy across four grids

Use the same two-layer interior support at every resolution.



p = 1.9965

Measured from the two finest grids.

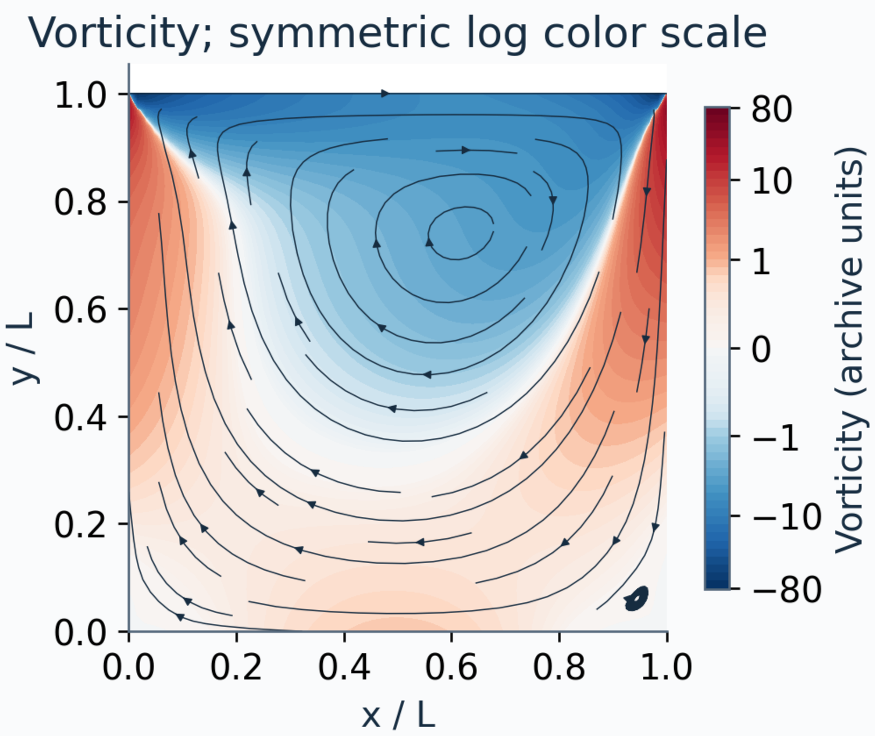
$$p = \frac{\log(E_{2h}/E_h)}{\log 2}$$

Finest-grid error: 4.015e-04
Grids: 17, 33, 65, 129

The measured rate agrees with the expected second-order stencil over this grid sequence.

Audit the retained Re = 100 cavity

The cavity has a moving lid; the manufactured field on the previous slides does not.



Grid

65 x 65

Interior divergence RMS

2.42e-16

Vorticity disagreement

2.902%

Wall maximum error

0.0e+00

The 2.902% value compares a derivative with archived vorticity; it is not error against exact flow.

Read the evidence against fixed thresholds

These tolerances belong to this float64 diagnostic and dataset.

Check	Observed	Required
Dataset identity	Exact SHA-256 match	Exact match
Observed convergence order	1.9965	At least 1.90
Finest analytic vorticity error	4.015e-04	At most 5e-4
Analytic divergence RMS	2.748e-14	At most 1e-12
Cavity divergence RMS	2.418e-16	At most 1e-12
Cavity/archive vorticity error	2.902e-02	At most 4e-2
Wall maximum error	0.0	At most 1e-12

All seven gates pass. A passed gate supports only the claim that the gate actually tests.

A plausible axis swap survives a shape test

Compare the correct vorticity with the intentionally wrong expression.

$$\omega_{\mathrm{wrong}} = D_y v - D_x u$$

Same shape. Finite values.
Different mathematical quantity.

The analytic reference detects the error before the cavity is used.

1.232

Relative L2 error

Required: at most 0.0005

Decision: REJECT

Keep this failing example: it shows what the acceptance procedure can detect.

Give the assistant a reviewable task

A scientific specification becomes the implementation brief.

Example implementation brief

Implement divergence and z-vorticity for u[y,x], v[y,x] on increasing Cartesian coordinates. Use second-order finite differences. Preserve the sign convention and the two-layer audit support. Return diagnostics; reject invalid inputs.

1. Propose

Code and tests

2. Execute

Frozen acceptance gates

3. Review

Physics, diff and evidence

Record AI assistance and human corrections. Scientific responsibility stays with the investigator.

Make the result reconstructable

The record must identify both the input and the reasoning behind acceptance.

Data identity

SHA-256 binds this result to the retained cavity archive. A hash mismatch rejects the record.

Execution context

Keep code revision, dependency versions and the exact command. Re-run from a clean checkout.

Decision evidence

Retain thresholds, measured values and failures. Inspect the record together with the implementation.

```
python qa/build_week01_1_materials.py --execute
```

Numerical reproducibility and byte-identical rendering are separate checks; both depend on environment.

Defend the claim with an evidence package

Submit the specification, implementation, test results and a short scientific review.

Assessment	Weight	Evidence
Specification	20%	Axes, units, sign, support and thresholds
Verification	25%	Analytic reference and measured convergence
Physical audit	20%	Boundaries and retained-field consistency
Manual review	15%	A scientific error found or ruled out
Provenance	10%	Input hash, revision, environment, command
Claim boundary	10%	Limits and retained rejection example

Discussion: what new reference and test are needed before extending this diagnostic to an unstructured mesh?

What the evidence supports

A verified Cartesian post-processor, with a consistency audit on one retained cavity case.

The next claim needs new evidence

Stretched meshes, unstructured grids and other boundary treatments require their own verification. Physical validation of a solver also requires an independent physical reference.

Wilson et al. (2017)	Good Enough Practices in Scientific Computing 10.1371/journal.pcbi.1005510
Sandve et al. (2013)	Ten Simple Rules for Reproducible Computational Research 10.1371/journal.pcbi.1003285
Roache (1998)	Verification and Validation in Computational Science and Engineering Book: Hermosa Publishers

Original FlowMLLab material. CS146S inspired the syllabus topic: themodernsoftware.dev